Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory (`https://assumptionsofphysics.org/resources/briefs`) were checked in Lean, where the formal version differs from the brief, and any problems found in the brief.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Superpositions correspond to multiple decompositions

Files.

- File: `briefs/SuperpositionAndMultipleMixtures/SuperpositionAndMultipleMixtures.lean`
- Library: `SuperpositionAndMultipleMixtures`
- Imports: `ForPhyslib.QuantumMechanics.Wigner.Density` (for `proj`, $|\psi\rangle\langle\psi|$, and its parallelogram law), `ForMathlib.Analysis.InnerProductSpace.Projection.FiniteDimensional`
- New `ForMathlib` material: `ForMathlib/Analysis/InnerProductSpace/Projection/FiniteDimensional.lean` (`Submodule.mem_span_of_forall_inner_eq_zero`: a vector orthogonal to everything that is orthogonal to a finite set lies in its span).

Setting. E is a complex inner product space of any dimension, not necessarily complete. For a vector ψ the pure state is `pureState` $\psi = \rho_\psi = |\psi\rangle\langle\psi| \,/\, \langle\psi|\psi\rangle$ (it is 0 at $\psi = 0$), and `mixture` $p\ \psi\ q\ \chi = p\rho_\psi + q\rho_\chi$. An operator is mixed (`IsMixed`) if it is not ρ_ψ for any $\psi \neq 0$. The conditions are:

- `QSupPur` $\phi\ \psi_1\ \psi_2$: ψ_1, ψ_2 linearly independent and $\phi = c_1\psi_1 + c_2\psi_2$ with $c_1, c_2 \neq 0$;
- `QSupMix` $\phi\ \psi_1\ \psi_2$: there are $p_1, p_2, p_\phi, p_{\hat\phi} \in (0,1)$ with $p_1 + p_2 = p_\phi + p_{\hat\phi} = 1$ and a state $\hat\phi \neq 0$ such that $\rho = p_1\rho_{\psi_1} + p_2\rho_{\psi_2}$ is mixed, $\rho = p_\phi\rho_\phi + p_{\hat\phi}\rho_{\hat\phi}$, and $\rho_\phi \notin \{\rho_{\psi_1}, \rho_{\psi_2}\}$.

Results.

Brief	Lean
eq. (2), $\|\phi\rangle\langle\phi\| + \|\hat\phi\rangle\langle\hat\phi\| = 2\|c_1\|^2\|\psi_1\rangle\langle\psi_1\| + 2\|c_2\|^2\|\psi_2\rangle\langle\psi_2\|$	`proj_add_proj_hat`
eqs. (3), (4), QSUP-PUR ⇒ QSUP-MIX	`QSupPur.qSupMix`, with the weights of eq. (3)
"it is supported by a two dimensional subspace"	`mem_span_of_mixture_eq`: if $q\rho_\phi + q'\rho_{\phi'} = p_1\rho_{\psi_1} + p_2\rho_{\psi_2}$ with $q > 0$, $q' \geq 0$, then $\phi \in \mathrm{span}\{\psi_1, \psi_2\}$
QSUP-MIX ⇒ QSUP-PUR	`QSupMix.qSupPur`, for $\phi \neq 0$
Theorem 1	`qSupPur_iff_qSupMix`, for $\phi \neq 0$
(used for QSUP-MIX)	`isMixed_mixture`: a mixture of two linearly independent states with positive weights is not pure

Differences from the brief.

- The equivalence is proved for each triple ϕ, ψ_1, ψ_2 with $\phi \neq 0$; no other vector needs to be nonzero, and E need not be complete or finite-dimensional.
- The brief writes ρ as a density operator. The formalization does not use the trace: ρ is a convex combination of two pure states, which is all the argument needs.
- QSUP-MIX ⇒ QSUP-PUR uses neither that ρ is mixed nor that $p_{\hat\phi} > 0$; only $p_\phi > 0$, $p_{\hat\phi} \geq 0$ and $\rho_\phi \notin \{\rho_{\psi_1}, \rho_{\psi_2}\}$.

Problems found in the brief. None that affect the theorem; the demonstration leaves out some steps, which the formalization supplies.

- QSUP-PUR ⇒ QSUP-MIX: the brief does not check that $\hat\phi \neq 0$ (needed to form $\rho_{\hat\phi}$), that $\rho_\phi \notin \{\rho_{\psi_1}, \rho_{\psi_2}\}$, or that ρ is mixed. All three follow from the linear independence of ψ_1, ψ_2 and $c_1, c_2 \neq 0$.
- QSUP-MIX ⇒ QSUP-PUR: "Since ρ is the mixture of two pure states, it is supported by a two dimensional subspace" presupposes that ψ_1 and ψ_2 are linearly independent, which QSUP-MIX does not state, and the brief does not show that $c_1, c_2 \neq 0$. Both follow from $\rho_\phi \notin \{\rho_{\psi_1}, \rho_{\psi_2}\}$: if ψ_2 were a multiple of ψ_1, or c_1 or c_2 were 0, then ϕ would be a multiple of ψ_1 or of ψ_2. The support claim itself is proved from the quadratic form: a vector orthogonal to ψ_1 and ψ_2 gives $\langle x|\rho|x\rangle = 0$, hence $\langle\phi|x\rangle = 0$.